

REAL-ROOTEDNESS OF A SUPER-RECURRENCE EULERIAN FAMILY

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ABSTRACT. We prove real-rootedness for a family of Eulerian-type polynomials introduced by U. Shankar. For an integer $l \geq 1$, the coefficients are defined by

$$E_{1,0}^{(l)} = 1, \quad E_{n,k}^{(l)} = (k+1)^l E_{n-1,k}^{(l)} + (n-k)^l E_{n-1,k-1}^{(l)}.$$

We show that the row polynomial $E_n^{(l)}(t) = \sum_{k=0}^{n-1} E_{n,k}^{(l)} t^k$ has only real non-positive zeros. In fact, we prove the stronger consecutive interlacing relation

$$E_n^{(l)}(t) \ll E_{n+1}^{(l)}(t)$$

for all $l \geq 1$ and $n \geq 1$. The proof is based on a Hadamard normalization of the rows, the Garloff–Wagner theorem on Hadamard products, and a small kernel interlacing argument.

1. INTRODUCTION

Eulerian polynomials are a central source of real-rooted and interlacing phenomena in enumerative combinatorics. One way to see this is through the classical recurrence

$$A_{n,k} = (k+1)A_{n-1,k} + (n-k)A_{n-1,k-1}.$$

Shankar [Sha25] considered a broad family of triangular arrays obtained by replacing the linear factors in such recurrences by powers. The Eulerian specialization is the following.

Definition 1.1. Fix $l \geq 1$. We define the *super-recurrence Eulerian coefficients* $E_{n,k}^{(l)}$ by

$$E_{1,0}^{(l)} = 1$$

and, for $n \geq 2$,

$$E_{n,k}^{(l)} = (k+1)^l E_{n-1,k}^{(l)} + (n-k)^l E_{n-1,k-1}^{(l)}, \quad (1.1)$$

where coefficients outside the range $0 \leq k \leq n-1$ are interpreted as zero. The *row polynomial* is

$$E_n^{(l)}(t) := \sum_{k=0}^{n-1} E_{n,k}^{(l)} t^k.$$

For $l = 1$ this is the usual Eulerian recurrence. For example, when $l = 2$ the first rows are

$$E_1^{(2)}(t) = 1, \quad E_2^{(2)}(t) = 1 + t, \quad E_3^{(2)}(t) = 1 + 8t + t^2,$$

Date: June 7, 2026.

2020 Mathematics Subject Classification. Primary 05A15; Secondary 26C10, 05A20.

Key words and phrases. Eulerian polynomial, real-rooted polynomial, interlacing, Hadamard product, Pólya frequency sequence.

and

$$E_4^{(2)}(t) = 1 + 41t + 41t^2 + t^3.$$

Shankar proved palindromicity and log-concavity properties for these arrays and recorded real-rootedness evidence. The main result of this note settles the real-rootedness question in the Eulerian specialization, and gives a stronger interlacing statement.

Theorem 1.2. *Let $l \geq 1$.*

- (a) *For every $n \geq 1$, the polynomial $E_n^{(l)}(t)$ has only real nonpositive zeros.*
- (b) *For every $n \geq 1$,*

$$E_n^{(l)}(t) \ll E_{n+1}^{(l)}(t).$$

Here $f \ll g$ denotes the usual weak proper-position relation for real-rooted polynomials, with the orientation fixed in Section 2. The proof also gives a refinement for a natural family of cumulative block-leader prefixes. We state it now and define the prefixes in Section 6.

Theorem 1.3. *For $l \geq 1$, $n \geq 1$, and $1 < m \leq n$, the cumulative block-leader prefixes satisfy*

$$P_{n,m-1}^{(l)}(t) \ll P_{n,m}^{(l)}(t).$$

Our proof has two main ideas. The first is a normalization. If $d = n - 1$, put

$$Q_n^{(l)}(t) := \sum_{i=0}^d \frac{E_{n,i}^{(l)}}{\binom{d}{i} t^i} t^i.$$

Then

$$E_n^{(l)}(t) = B_{d,l}(t) \odot Q_n^{(l)}(t), \quad B_{d,l}(t) = \sum_{i=0}^d \binom{d}{i}^l t^i,$$

where $\odot$ denotes coefficientwise Hadamard product. The kernel $B_{d,l}$ is the l -fold Hadamard power of $(1+t)^d$, while the normalized rows $Q_n^{(l)}$ satisfy a recurrence built from the two operators $N - \theta$ and $\theta + 1$, where $\theta = t d/dt$. These operators preserve the cone of polynomials with nonnegative coefficients and only nonpositive real zeros. This proves part (a) of Theorem 1.2.

The second idea is that the same normalization is compatible with interlacing. We use the two-pair form of the Garloff–Wagner Hadamard theorem [GW96, Thm. 4]. A small pair of explicit kernels,

$$V_d(t) = (1+t)^d, \quad U_d(t) = \sum_{i=0}^d (i+1) \binom{d}{i} t^i = (1+t)^{d-1} (1+(d+1)t),$$

contains all the needed root-list information. Hadamard powers of the relations among V_d , U_d , and the shifted reciprocal of U_d then yield the consecutive interlacing in part (b).

2. REAL-ROOTEDNESS TOOLS

We collect the standard facts about real-rooted polynomials that are used in the proof. For background on Pólya frequency sequences, real-rootedness, compatibility, and interlacing, see Aissen–Edrei–Schoenberg–Whitney [AESW51], Brenti [Bre89], Chudnovsky–Seymour [CS07], and Brändén [Brä15].

Definition 2.1. A polynomial $f \in \mathbb{R}[t]$ is a *PF polynomial* if all its coefficients are nonnegative and all its zeros are real and nonpositive. We use the convention that the zero polynomial is allowed when it appears as a limiting or degenerate case.

If $f, g \in \mathbb{R}[t]$ have only real zeros, we write $f \ll g$ if the zeros of f weakly interlace or weakly alternate to the left of the zeros of g , in the following orientation. If $\deg g = \deg f + 1$, the roots can be listed as

$$\beta_1 \leq \alpha_1 \leq \beta_2 \leq \cdots \leq \alpha_{\deg f} \leq \beta_{\deg g},$$

where the α_i are roots of f and the β_i are roots of g . If $\deg f = \deg g$, the roots can be listed as

$$\alpha_1 \leq \beta_1 \leq \alpha_2 \leq \cdots \leq \alpha_{\deg f} \leq \beta_{\deg g}.$$

We use the closed weak form, so repeated roots and zero polynomials are handled by the usual limiting convention.

For $f(t) = \sum_i a_i t^i$ and $g(t) = \sum_i b_i t^i$, define their *Hadamard product*

$$(f \odot g)(t) := \sum_i a_i b_i t^i,$$

where coefficients are extended by zero outside the degree range. We also write

$$\mathcal{I}_D f(t) := t^D f(1/t)$$

for the *shifted reciprocal*.

Theorem 2.2 (Schur–Pólya–Wagner; Garloff–Wagner). *Let $f, g, p, q \in \mathbb{R}[t]$ be PF polynomials.*

- (a) *If f and p are PF, then $f \odot p$ is PF.*
- (b) *If $f \ll g$ and $p \ll q$, then*

$$f \odot p \ll g \odot q.$$

Part (a) is the classical Schur–Pólya–Wagner preservation theorem, in the nonnegative-coefficient convention used here. Part (b) is the two-pair proper-position form of Garloff–Wagner [GW96, Thm. 4]; see also the earlier work of Schur and Pólya [SP14].

Theorem 2.3 (Fixed-left cone property). *Let $p, q, r \in \mathbb{R}[t]$ be PF polynomials. If $p \ll q$ and $p \ll r$, then*

$$p \ll aq + br$$

for all $a, b \geq 0$.

This is the standard convexity of a common interlacing cone. It is equivalent to the two-generator compatibility criterion of Obreschkoff type; in the form used here it is also contained in the addition lemmas used by Garloff–Wagner [GW96, Lem. 6].

We shall use two elementary differential operators. Let

$$\theta f(t) := t f'(t).$$

Lemma 2.4. *Let $f \in \mathbb{R}[t]$ be PF.*

- (a) *The polynomial $(\theta + 1)f = (tf)'$ is PF.*
- (b) *If $\deg f \leq N$, then $(N - \theta)f$ is PF.*

Proof. The coefficients are plainly nonnegative in both cases, in the second case because the coefficient of t^i in $(N - \theta)f$ is $(N - i)[t^i]f$.

For the roots, part (a) follows from Rolle's theorem applied to tf . Indeed, the polynomial tf has only real nonpositive zeros, and its derivative is $(\theta + 1)f$.

For part (b), remove the zero roots of f . Thus write

$$f(t) = t^s h(t), \quad h(t) = c \prod_{j=1}^m (t + a_j),$$

where $c > 0$ and $a_j > 0$. Since $s + m = \deg f \leq N$, we have

$$(N - \theta)f = t^s ((N - s - \theta)h).$$

Moreover,

$$(N - s - \theta)h = (N - s - m)h + \sum_{j=1}^m a_j \frac{h}{t + a_j}.$$

The polynomials $h/(t + a_j)$ weakly interlace h , and all coefficients in this linear combination are nonnegative. By Theorem 2.3, the right-hand side is again real-rooted with only nonpositive zeros. Multiplying by t^s only appends zero roots. This completes the proof. $\square$

Lemma 2.5. *The operator $\theta + 1$ preserves weak proper position on the PF cone. In particular, if f and g are PF and $f \ll g$, then*

$$(\theta + 1)f \ll (\theta + 1)g.$$

Consequently, every power $(\theta + 1)^r$ has the same property.

Proof. The compatibility characterization of proper position says that $f \ll g$ is equivalent, in this nonnegative PF setting, to real-rootedness of all nonnegative linear combinations $af + bg$. Applying Lemma 2.4(a) to each such combination gives that

$$a(\theta + 1)f + b(\theta + 1)g$$

is PF for all $a, b \geq 0$. The same characterization gives the desired proper-position relation. Iterating proves the final assertion. $\square$

3. THE NORMALIZED ROWS

The proof of Theorem 1.2 starts with a normalization which removes the binomial part of the coefficients.

Definition 3.1. For $d = n - 1$, define

$$Q_n^{(l)}(t) := \sum_{i=0}^d \frac{E_{n,i}^{(l)}}{\binom{d}{i} t^i} t^i.$$

We also define

$$B_{d,l}(t) := \sum_{i=0}^d \binom{d}{i}^l t^i.$$

Lemma 3.2 (Normalized transfer identity). *For $l \geq 1$ and $n \geq 2$,*

$$Q_n^{(l)}(t) = (1 + t) \left(\frac{(\theta + 1)(n - 1 - \theta)}{n - 1} \right)^l Q_{n-1}^{(l)}(t).$$

Proof. Put $N = n - 1$ and write

$$Q_{n-1}^{(l)}(t) = \sum_{k=0}^{N-1} q_k t^k.$$

The operator

$$\frac{(\theta + 1)(N - \theta)}{N}$$

is diagonal in the monomial basis: it sends t^k to

$$\frac{(k + 1)(N - k)}{N} t^k.$$

Thus, after applying the l th power and multiplying by $1 + t$, the coefficient of t^k is

$$\left(\frac{(k + 1)(N - k)}{N} \right)^l q_k + \left(\frac{k(N + 1 - k)}{N} \right)^l q_{k-1}, \quad (3.1)$$

where $q_{-1} = q_N = 0$.

Since

$$q_k = \frac{E_{n-1,k}^{(l)}}{\binom{N-1}{k}^l},$$

the first summand in (3.1) is

$$\frac{(k + 1)^l E_{n-1,k}^{(l)}}{\binom{N}{k}^l},$$

using

$$\binom{N}{k} = \frac{N}{N - k} \binom{N - 1}{k}.$$

Similarly, the second summand is

$$\frac{(N + 1 - k)^l E_{n-1,k-1}^{(l)}}{\binom{N}{k}^l},$$

using

$$\binom{N}{k} = \frac{N}{k} \binom{N - 1}{k - 1}.$$

Since $N + 1 = n$, the sum is

$$\frac{(k + 1)^l E_{n-1,k}^{(l)} + (n - k)^l E_{n-1,k-1}^{(l)}}{\binom{n-1}{k}^l} = \frac{E_{n,k}^{(l)}}{\binom{n-1}{k}^l}.$$

This is exactly the coefficient of t^k in $Q_n^{(l)}$. □

Lemma 3.3. *For all $l \geq 1$ and $n \geq 1$, the polynomial $Q_n^{(l)}$ is PF.*

Proof. We proceed by induction on n . The base case is $Q_1^{(l)} = 1$. Assume that $Q_{n-1}^{(l)}$ is PF. Its degree is at most $n - 2$. Therefore Lemma 2.4(b), with $N = n - 1$, shows that $(n - 1 - \theta)Q_{n-1}^{(l)}$ is PF. Lemma 2.4(a) then shows that applying $\theta + 1$ preserves the PF property. The scalar factor $1/(n - 1)$ is positive, and multiplication by $1 + t$ only adds the root -1 . Iterating this l times in Lemma 3.2 proves that $Q_n^{(l)}$ is PF. □

Lemma 3.4. For $d = n - 1$,

$$E_n^{(l)} = B_{d,l} \odot Q_n^{(l)}.$$

Consequently, $E_n^{(l)}$ is PF for all $l \geq 1$ and $n \geq 1$.

Proof. The coefficient of t^i in $B_{d,l} \odot Q_n^{(l)}$ is

$$\binom{d}{i}^l \frac{E_{n,i}^{(l)}}{\binom{d}{i}^l} = E_{n,i}^{(l)}.$$

This proves the factorization.

Moreover, $B_{d,l}$ is the l -fold Hadamard power of $(1+t)^d$, since

$$(1+t)^d = \sum_{i=0}^d \binom{d}{i} t^i.$$

The polynomial $(1+t)^d$ is PF, and so Theorem 2.2(a) implies that $B_{d,l}$ is PF. By Lemma 3.3, the other factor $Q_n^{(l)}$ is PF. Applying Theorem 2.2(a) once more proves that $E_n^{(l)}$ is PF. $\square$

This proves Theorem 1.2(a).

4. PALINDROMICITY AND THE HADAMARD KERNELS

We now prepare the interlacing proof. We first record the palindromicity, which is also proved in Shankar's work [Sha25].

Lemma 4.1. For $0 \leq k \leq n - 1$,

$$E_{n,k}^{(l)} = E_{n,n-1-k}^{(l)}.$$

Consequently, both $E_n^{(l)}$ and $Q_n^{(l)}$ are palindromic of degree $n - 1$.

Proof. We proceed by induction on n . The claim is immediate for $n = 1$. Assume it for row $n - 1$. Applying the recurrence to the index $n - 1 - k$ gives

$$E_{n,n-1-k}^{(l)} = (n-k)^l E_{n-1,n-1-k}^{(l)} + (k+1)^l E_{n-1,n-2-k}^{(l)}.$$

By the induction hypothesis in row $n - 1$,

$$E_{n-1,n-1-k}^{(l)} = E_{n-1,k-1}^{(l)} \quad \text{and} \quad E_{n-1,n-2-k}^{(l)} = E_{n-1,k}^{(l)},$$

with the usual zero convention at the boundary. Hence

$$E_{n,n-1-k}^{(l)} = (n-k)^l E_{n-1,k-1}^{(l)} + (k+1)^l E_{n-1,k}^{(l)} = E_{n,k}^{(l)}.$$

The assertion for $Q_n^{(l)}$ follows because $\binom{n-1}{k} = \binom{n-1}{n-1-k}$. $\square$

Define the second Hadamard kernel by

$$K_{d,l}(t) := \sum_{i=0}^d (i+1)^l \binom{d}{i}^l t^i.$$

We shall compare $B_{d,l}$, $K_{d,l}$, and the shifted reciprocal $\mathcal{I}_{d+1} K_{d,l}$.

Lemma 4.2. For all $d \geq 0$ and $l \geq 1$,

$$B_{d,l} \ll K_{d,l}, \quad B_{d,l} \ll \mathcal{I}_{d+1} K_{d,l}, \quad K_{d,l} \ll \mathcal{I}_{d+1} K_{d,l}.$$

Proof. The case $d = 0$ is immediate, so assume $d \geq 1$. Put

$$V_d(t) = (1 + t)^d$$

and

$$U_d(t) = \sum_{i=0}^d (i+1) \binom{d}{i} t^i.$$

Since

$$U_d(t) = (\theta + 1)V_d(t) = (t(1+t)^d)' = (1+t)^{d-1}(1 + (d+1)t),$$

the roots of U_d are -1 , with multiplicity $d-1$, and $-1/(d+1)$. The polynomial $\mathcal{I}_{d+1}U_d$ has roots 0 , $-(d+1)$, and -1 with multiplicity $d-1$. Comparing these ordered root lists gives

$$V_d \ll U_d, \quad V_d \ll \mathcal{I}_{d+1}U_d, \quad U_d \ll \mathcal{I}_{d+1}U_d.$$

Now

$$B_{d,l} = V_d^{\odot l}, \quad K_{d,l} = U_d^{\odot l}, \quad \mathcal{I}_{d+1}K_{d,l} = (\mathcal{I}_{d+1}U_d)^{\odot l}.$$

The last identity follows coefficientwise from the definition of the shifted reciprocal. Repeated application of Theorem 2.2(b) to the three displayed interlacings gives the result. $\square$

5. CONSECUTIVE INTERLACING

We now prove the stronger part of Theorem 1.2. The proof is a direct decomposition of the two summands in the recurrence for $E_{n+1}^{(l)}$.

Lemma 5.1. *Let $n \geq 1$, put $d = n - 1$, and set*

$$p(t) = E_n^{(l)}(t), \quad A(t) = (\theta + 1)^l p(t).$$

Then

$$E_{n+1}^{(l)}(t) = A(t) + t^n A(1/t).$$

Proof. The recurrence for row $n + 1$ gives

$$E_{n+1}^{(l)}(t) = ((\theta + 1)^l + t(n - \theta)^l)p(t).$$

The first summand is $A(t)$. We show that the second summand is $t^n A(1/t)$. Write

$$p(t) = \sum_{i=0}^d a_i t^i.$$

By Lemma 4.1, $a_i = a_{d-i}$. The coefficient of t^j in $t(n - \theta)^l p(t)$ is

$$(n - j + 1)^l a_{j-1}$$

for $1 \leq j \leq n$, and it is zero for $j = 0$. On the other hand, the coefficient of t^j in $t^n A(1/t)$ is

$$(n - j + 1)^l a_{n-j} = (n - j + 1)^l a_{j-1},$$

again for $1 \leq j \leq n$. Thus the two polynomials agree. $\square$

Lemma 5.2. *With the notation of Lemma 5.1,*

$$p = B_{d,l} \odot Q_n^{(l)}, \quad A = K_{d,l} \odot Q_n^{(l)},$$

and

$$t^n A(1/t) = (\mathcal{I}_{d+1}K_{d,l}) \odot tQ_n^{(l)}.$$

Proof. The first identity is Lemma 3.4. For the second, the coefficient of t^i in $A = (\theta + 1)^l p$ is

$$(i + 1)^l E_{n,i}^{(l)} = (i + 1)^l \binom{d}{i}^l [t^i] Q_n^{(l)},$$

which is the coefficient of t^i in $K_{d,l} \odot Q_n^{(l)}$.

For the last identity, compare coefficients. Since $n = d + 1$, the coefficient of t^j in the right-hand side is

$$(d + 2 - j)^l \binom{d}{d + 1 - j}^l [t^{j-1}] Q_n^{(l)}.$$

Using

$$\binom{d}{d + 1 - j} = \binom{d}{j - 1}$$

and palindromicity of $Q_n^{(l)}$, this is exactly the coefficient of t^j in $t^n A(1/t)$. $\square$

Proof of Theorem 1.2(b). For $n = 1$, the assertion is $1 \ll 1 + t$, which is immediate. Assume $n \geq 2$ and use the notation of Lemma 5.1. By Lemma 5.2,

$$p = B_{d,l} \odot Q_n^{(l)}, \quad A = K_{d,l} \odot Q_n^{(l)}.$$

The normalized row $Q_n^{(l)}$ is PF by Lemma 3.3, so

$$Q_n^{(l)} \ll Q_n^{(l)}.$$

Together with $B_{d,l} \ll K_{d,l}$ from Lemma 4.2, the two-pair Hadamard theorem gives

$$p \ll A.$$

Similarly,

$$t^n A(1/t) = (\mathcal{I}_{d+1} K_{d,l}) \odot t Q_n^{(l)}.$$

Since $Q_n^{(l)}$ is PF, the root list immediately gives

$$Q_n^{(l)} \ll t Q_n^{(l)}.$$

Combining this with

$$B_{d,l} \ll \mathcal{I}_{d+1} K_{d,l}$$

and applying Theorem 2.2(b), we get

$$p \ll t^n A(1/t).$$

The fixed-left cone property, Theorem 2.3, now gives

$$p \ll A + t^n A(1/t).$$

By Lemma 5.1, the right-hand side is $E_{n+1}^{(l)}$. This proves

$$E_n^{(l)} \ll E_{n+1}^{(l)},$$

as desired. $\square$

The same proof gives a reciprocal endpoint which will be useful for the prefix refinement.

Corollary 5.3. *Let*

$$A_n^{(l)}(t) := (\theta + 1)^l E_n^{(l)}(t).$$

Then

$$A_n^{(l)}(t) \ll t^n A_n^{(l)}(1/t)$$

and

$$A_n^{(l)}(t) \ll E_{n+1}^{(l)}(t).$$

Proof. The first statement follows from the same Hadamard decomposition as above, using

$$K_{d,l} \ll \mathcal{I}_{d+1} K_{d,l} \quad \text{and} \quad Q_n^{(l)} \ll t Q_n^{(l)}.$$

The second follows from the fixed-left cone property and

$$E_{n+1}^{(l)}(t) = A_n^{(l)}(t) + t^n A_n^{(l)}(1/t).$$

This completes the proof. $\square$

6. CUMULATIVE BLOCK-LEADER PREFIXES

We finish with the prefix refinement. We use a block-leader product formula for the rows. A *block-leader set* is a subset $B \subseteq [n]$ with $1 \in B$. For such a set, define

$$c_n(B) := \prod_{i=1}^n w_i(B),$$

where, if

$$r_i = |B \cap [i-1]|,$$

then

$$w_i(B) = \begin{cases} i - r_i, & i \in B, \\ r_i, & i \notin B. \end{cases}$$

Equivalently, if $B = \{1 = b_1 < b_2 < \dots < b_r\}$ and $b_{r+1} = n+1$, then

$$c_n(B) = \prod_{j=1}^r (b_j - j + 1) \prod_{j=1}^r j^{b_{j+1} - b_j - 1}.$$

Lemma 6.1. *We have*

$$E_n^{(l)}(t) = \sum_{\substack{B \subseteq [n] \\ 1 \in B}} c_n(B) t^{|B|-1}.$$

Proof. Let the right-hand side be $F_n^{(l)}(t)$. We verify the same recurrence and initial condition as in Definition 1.1. The case $n = 1$ is clear.

Fix $n \geq 2$ and split the block-leader sets according to whether n belongs to B . If $n \notin B$ and $|B| = k+1$, then B is a block-leader set in $[n-1]$ and

$$c_n(B) = (k+1) c_{n-1}(B).$$

These terms contribute

$$(k+1)^l E_{n-1,k}^{(l)}$$

to the coefficient of t^k .

If $n \in B$, then $B' = B \setminus \{n\}$ is a block-leader set in $[n-1]$ with $|B'| = k$. In this case

$$c_n(B) = (n-k) c_{n-1}(B'),$$

and the contribution to the coefficient of t^k is

$$(n-k)^l E_{n-1,k-1}^{(l)}.$$

Thus the coefficients of $F_n^{(l)}$ satisfy (1.1). The claim follows. $\square$

Definition 6.2. For $1 \leq m \leq n$, define the *cumulative block-leader prefix*

$$P_{n,m}^{(l)}(t) := \sum_{\substack{B \subseteq [n] \\ 1 \in B, \max B \leq m}} c_n(B) t^{|B|-1}.$$

Note that $P_{n,n}^{(l)} = E_n^{(l)}$ by Lemma 6.1.

Lemma 6.3. For $1 \leq m \leq n$,

$$P_{n,m}^{(l)}(t) = (\theta + 1)^{l(n-m)} E_m^{(l)}(t).$$

Proof. If $B \subseteq [m]$ and we view B as a subset of $[n]$, then every index $i > m$ is not in B and has $r_i = |B|$. Hence

$$c_n(B) = |B|^{n-m} c_m(B).$$

Using Lemma 6.1, we get

$$P_{n,m}^{(l)}(t) = \sum_{\substack{B \subseteq [m] \\ 1 \in B}} |B|^{l(n-m)} c_m(B) t^{|B|-1}.$$

The operator $\theta + 1$ multiplies the monomial $t^{|B|-1}$ by $|B|$. Therefore $(\theta + 1)^{l(n-m)}$ gives exactly the displayed factor $|B|^{l(n-m)}$. This proves the identity. $\square$

Proof of Theorem 1.3. It is enough to prove that

$$P_{n,m}^{(l)} \ll P_{n,m+1}^{(l)} \quad (m < n),$$

and then replace m by $m - 1$. By Corollary 5.3,

$$(\theta + 1)^l E_m^{(l)} \ll E_{m+1}^{(l)}.$$

Apply the operator

$$(\theta + 1)^{l(n-m-1)}$$

to both sides. This preserves weak proper position by Lemma 2.5. Using Lemma 6.3, the resulting relation is exactly

$$P_{n,m}^{(l)} \ll P_{n,m+1}^{(l)}.$$

The claim follows. $\square$

Remark 6.4. The normalized gamma-polynomial refinements suggested by computations are not used in the proof above. The Hadamard normalization and the small kernel interlacing argument are sufficient for the real-rootedness and consecutive interlacing statements.

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